

WEIGHT MANAGEMENT



Mathieu Fontaine stalking *Predator*, 5.13b, Rumney, NH.

 Lee Hansche

CHAPTER

11

*"Nothing tastes as good as skinny feels."
— Supermodel Kate Moss³²*

THE CASE FOR CLIMBERS TO MANAGE WEIGHT

Controlling body weight is critical to maximizing climbing performance. Climbers go to great lengths to develop finger and upper-body strength, with the aim of improving their strength-to-weight ratio in order to resist the pull of gravity. They work very hard

(and proud) on the "strength" side of that equation, so it would be foolish to ignore the "weight" side. For those still not certain of the relevance of body weight, a season's worth of quantified hangboard training should demonstrate the difference 10 pounds makes when hanging from a small hold. From a purely biomechanical standpoint, the steeper the climbing, the more significant body weight becomes — steeper climbs require the upper body to support more of the climber's weight. However, less-steep routes at one's limit will have small holds with limited capacity to support the climber's weight, so even slightly slabby to vertical routes will be easier for leaner climbers.

Do I Have To?

Don't be ashamed if you're not excited about this topic: Weight loss and dieting aren't fun (as former wrestlers, both Mark and I have seen the drawbacks of performance-motivated weight loss, and cringe at the thought of dieting). Weight management is a tolerable sacrifice, however, one of many you already make for climbing. It becomes more tolerable with these two realizations:

- 1) It is very effective at improving climbing performance
- 2) With periodic training, it's not necessary to deprive yourself constantly.

A "performance diet" can be periodic, just like training, producing periods of samurai-like self-denial and periods of relaxation, celebration, and all-out gluttony! Just like other physical attributes (strength, endurance, etc.), low weight cannot be maintained indefinitely, so it makes sense to cycle it up and down with your training. Many high-performance athletes do this in other sports, including professional cyclists and bodybuilders. The latter group in particular needs to be very lean for competitions, to achieve the "cut" look, but muscle builds more quickly with more body fat, so they bulk up during mass-building phases then work off the fat in the weeks leading up to the competition, precisely timing their diet to appear "sliced" in time to impress the judges.

This approach provides a break from the stress and self-denial of maximizing performance while maintaining the benefits. Once the goal weight is reached, it's much easier to maintain it throughout the Performance Phase due in part to the added time spent at the crag. Dieting is also easier knowing that the self-deprivation is only temporary, and aimed at a motivating goal with a finite time limit. When the goal is achieved, you can return to some of those vices that make life worth living. To say it plainly, it's much easier to go without when you know it's not permanent.

— Mike Anderson

"I spent time with [Wolfgang Gullich] in 1991 and he told me he was trying a new climb, and although he made no mention of its difficulty, I could see it was pushing him. He was dieting... he was incredibly ripped. The skin around his stomach was like the skin on the back of your hand." — Jerry Moffatt¹⁸

HOW LEAN, AND WHEN TO GET THERE

Every body is unique, and some stay lean more easily than others. For many climbers (especially older ones), weight management can be very difficult, but it offers tremendous return on the investment. A climber might spend six years to increase his two-finger pocket strength by 15 pounds, or (depending on his body composition) he could lose 15 pounds in six weeks and achieve the same climbing-performance benefits! Of course, the potential for such weight loss is limited. There is a point of diminishing returns where the benefit of losing a few extra pounds through extreme calorie restriction does not make up for the resulting reduction in muscle mass and general lack of energy.

Athletes should strive to be generally fit, with body fat less than 10 percent for men and 20 percent for women. In most cases, exceeding these thresholds places excessive stress on the joints, increasing the risk of injury and reducing enthusiasm for exercise. During the Performance Phase of a training cycle, when striving to maximize performance, it is beneficial to *temporarily* trim down to as low as 5 to 6 percent body fat for men, and 12 to 14 percent for women.^{12, 30} (Body-fat percentage is difficult to measure in practice; therefore, find your optimal climbing weight by tracking performance, as described later in this chapter.) Each climber may need to experiment to determine where that point of diminishing returns is, or how lean he or she can be before the low-energy malaise that comes with dieting begins to overpower any strength-to-weight ratio benefits achieved. Record your body weight in a training journal, along with notes about the corresponding energy level and quality of performance, in order to determine an ideal "fighting weight."

At the end of each season's peak, it is acceptable (and even desirable), to relax dietary restrictions and bulk up by five to ten pounds throughout the Rest Phase and into the following Strength Phase. It is very difficult to add muscle and effectively build strength

with restricted caloric intake, because it is very hard to determine the correct number of calories to consume each day, as well as the correct ratio of carbs, fat, and protein. Undereating after a workout will inhibit the body's ability to restore glycogen stores in the muscles, leading to reduced performance in subsequent workouts. Worse, insufficient nutrition could weaken connective tissue, increasing risk of injury. Therefore, it's best not to attempt dieting through the Strength Phase. Instead, a climber in training should eat freely through the Strength Phase, and start dropping weight during the Power Phase. Weight loss can continue into the Performance Phase, but the resulting lethargy may hinder performance. It's better to reach the goal weight before the Performance Phase, then increase consumption slightly to enjoy higher energy levels throughout the remainder of the season.



Low body fat is an advantage on difficult climbs, but excessive dieting can sap strength and lead to injury or illness. To determine your optimal "send weight", experiment and note the results.  Mike Anderson

Weight Loss Is Not Cumulative

Body weight and strength are both closely related to climbing performance, but they have different characteristics. Though we cycle strength on and off through periodic training, it gradually builds over time. The work put in five to ten years ago contributes to your current level of strength, which is to say, strength training is cumulative over your lifetime, to an extent. Weight loss is not. The weight I lost in high school does nothing for me today. Therefore, if life's circumstances prevent you from climbing for an extended period, there is little long-term performance benefit to staying ultra-lean. On the other hand, you would be wise to continue strength training, because the strength gained during your hiatus will contribute to your strength when you return. For example, from 2005 to 2009, I was sidetracked by the adventures of parenthood, severely limiting my climbing time. Therefore, any effort to stay lean would have gone unrewarded. Nevertheless, I kept training. Once my kids were old enough for us to climb more, I was able to trim down to maximize performance, and I wasn't penalized for the years that I "let myself go." Once I lost the baby weight, I achieved new career bests in redpoint and onsight ability.

— Mike Anderson



The author in chubbier times, in the midst of a PhD program and raising young children (left), and two years later, 25 pounds lighter, and two letter grades harder (right). (left) *Rude Boys*, 5.13b, Smith Rock, OR: Mark Anderson; (right) *True Love*, 5.13d, RRG, KY: Anthony Carco

THE FUNDAMENTAL DIET

RULE: THE CALORIC DEFICIT

With the requirement for weight management firmly established, the question of how best to achieve it comes into focus. The merits of various weight-loss programs have been debated endlessly, with new “miracle” diets appearing on the scene regularly — low-fat, Atkins, low-carb, Paleo, Zone, South Beach, the list goes on and on. It can be confusing trying to sort through the pros and cons of these diets, and very difficult to reconcile the misleading marketing with sincere testimonials from trusted sources. To make matters worse, rigorous scientific evaluation of these diets is inconsistent. A few programs receive lots of attention from the scientific community (low-fat and Atkins, especially) while the others remain unevaluated.² Even when they have been rigorously studied, the results are often inconclusive or contradictory.

The basis for many of these diets is an oversimplified explanation of the body’s metabolic processes, which is then wildly extrapolated to create seductive theories about how the body might respond to certain combinations of food. The result is a convincing explanation for why such-and-such diet will make you shed pounds with ease, which is really a hollow house of cards built on a foundation of assumptions and outright speculation. Without rigorous scientific study, any claims of efficacy should be viewed with ample skepticism. Among all of the false claims and pseudoscience trolling the diet world, there is one universally applicable and indisputable fact: If more calories are burned than are consumed, weight goes down. Vice versa, and weight goes up. There is no way around this fundamental principle. No matter the chosen diet plan, if this *caloric deficit* is accomplished, body weight will decrease. Most climbers should aim to consume 1,500-1,800 calories per day while dieting — depending on body size and metabolism, but it shouldn’t be necessary to meticulously count calories, as discussed in Baseline Diet, below.

The traditional, “blue collar” method of dieting is to eat smaller portions of generally “healthy” foods (i.e., low fat, low sugar, and low sodium). This method is tried and true, and effective, but it can also be very difficult for many people, as it requires considerable willpower. These diets only work if close attention is paid to food labels and the nutritional contents are correctly identified (especially challenging when eat-

ing out). Otherwise it is nearly impossible to know if calories are being reduced. Just because something is described as “low fat” doesn’t mean it is healthy, or will cause weight loss. Low fat does not mean low calorie! One problem with “low-fat” diets (which were all the rage through the 1970s, ‘80s, and ‘90s) is that when the food engineers set out to reduce fat in foods, they replaced it with sugar and other carbs, which lack the *satiety* (feeling of satisfaction) of fat, despite the fact that the body easily converts them into fat for long-term storage.¹ Such engineered low-fat foods cause blood-sugar spikes, which are worse than eating the fat in the first place, because they disrupt how food energy is burned, consumed, and stored by the body.

SKEWING THE CALORIC EQUATION TOWARD WEIGHT LOSS

Many of the recent innovative diets (such as Atkins, South Beach, and Paleo, among others) seek to control weight by accounting for the speed at which digested carbs affect blood sugar. Despite the marketing tag lines such as “you can eat until you’re full” and “no counting calories,” these diets still depend on the fundamental principle of caloric deficit to stimulate fat loss. Therefore, these diets still require personal restraint and discipline. However, they differ from the “blue collar” diet in that they seek to skew the fundamental caloric equation in the dieter’s favor, and thus lessen the requirement for restraint and discipline. This is done by attempting to manipulate key processes in the body that control:

1. Metabolism
2. Appetite
3. Fat storage

These processes are fundamental to how the body digests food, converting it to usable energy. These control mechanisms are used by the body to maintain blood sugar at an optimal level as food is consumed.⁵ However, climbers can use these control mechanisms in reverse to control their weight. *By carefully selecting the foods eaten*, metabolism, appetite, and fat storage can be controlled, making it *easier to achieve a caloric deficit and thus easier to diet*.²⁸ These dietary techniques help to:

- Increase metabolism — allowing dieters to burn calories faster
- Reduce appetite — making it easier to eat less
- Manipulate the mechanism for fat storage — reducing the body’s calorie retention

These effects can be achieved by influencing the body's glycemic response.

Glucose present in the blood (*blood glucose*, or *blood sugar*) is the body's preferred energy source for muscular activity and brain function (as opposed to glycogen stored locally in the muscles, or energy stored as fat in cells). In fact, glucose is the sole energy source for brain function, and since the brain is the most important organ, the human body has a robust system for maintaining adequate levels of blood sugar called the *glycemic response*. Glucose is provided to the blood by carbohydrates, and some carbs digest more quickly than others. When blood sugar is low, the body craves carbs and attempts to conserve energy (resulting in lethargic feelings). If carbohydrates are not ingested, the body will initiate the process of *ketosis*, which breaks down stored fat into fatty acids that can be used as energy in place of glucose (except

in the brain).¹⁰ Therefore, some low-carb diets claim that this induced state of ketosis forces the consumption of unsightly stored fat — however, these claims remain controversial and generally unverified.² In fact, any diet that reduces fat must induce ketosis, but low-carb diets seek to do so without extreme calorie restriction.

When blood sugar is high, our pancreas releases insulin, which signals the cells to absorb and store glucose, thus lowering blood sugar. Muscle and liver tissue store glucose by converting it to glycogen, up to a point — the rest is used to convert fatty acids to stored fat. The body seeks to control blood sugar in a narrow, optimal range. Therefore, when a rapid increase is detected, the body overcompensates, releasing excess insulin and triggering the rapid absorption of glucose by the cells. This is helpful immediately after a workout, when local muscular glycogen has been depleted and needs to be replaced quickly to facilitate recovery. However, at any other time of day, it is undesirable.

This rapid absorption of glucose will quickly saturate the muscles' capacity to store it as glycogen, so most of it will be stored as fat. Furthermore, the body can remove this excess glucose from the blood quite rapidly, lowering the blood sugar below optimal levels before the pancreas has time to shut off insulin production. This, ironically, causes a *low-blood-sugar* state and a corresponding *lowering* of metabolism to conserve energy (the "sugar crash"), and induces cravings for more high-sugar carbs. If those cravings are consummated, the result is a vicious cycle of rapidly rising and lowering blood sugar that leads to reduced metabolism, increased appetite, and excess fat storage. This trifecta has a devastating effect on the caloric equation.

Ideally, blood sugar should be maintained at a relatively constant level, with only small fluctuations that occur slowly over many hours. This gives the pancreas ample time to detect changes and react with measured insulin production. Rapid, massive spikes in blood sugar overwhelm the pancreas' ability to control blood-sugar levels, and lead to the aforementioned vicious cycle. Therefore, many dietitians have reasoned that foods with a low *glycemic index* (GI) — i.e., those that require more time to break down into blood glucose — may be beneficial to weight management.^{4, 21}



The Mistress of Fitness Alli Rainey, lean and mean on F'd in the A, 5.14a, Tensleep, WY  Andrew Burr

Recovery Plan

Proper post-workout nutrition is critical for climbers, because of the reliance on local glycogen in the forearms and the need to stay lean. For up to two hours immediately after a workout, carbs are converted to muscular glycogen three times faster than normal.^{3, 14} Therefore, carbs ingested during this period are more likely to end up as useful muscle glycogen rather than stored fat (so it's OK to eat high-GI foods during this two-hour window). Further, consuming foods with a 4:1 ratio (by weight) of carbs to protein is known to improve the glycogen storage process.³ Quickly replenishing muscle glycogen sets you up for your next workout or climbing day.¹⁴ That said, climbing doesn't exhaust the largest muscle groups, so massive glycogen loading (like that done by endurance athletes) is not required for climbing.

After intense strength-training workouts (hangboard, campusing, or bouldering sessions), I typically eat half of a Clif Builder's protein bar and a banana, and drink at least 16 ounces of water — total nutritional value: 42g carbs, 10g protein, 240 calories. This provides the perfect recovery ratio and plenty of calories for a climbing workout. (Many other products are available with optimal recovery formulations.) At the crag, I nibble on snacks after every burn, typically eating an apple, a banana, a protein bar, two to three granola or energy bars, and two liters of water throughout the day.

— Mike Anderson

GLYCEMIC INDEX AND GLYCEMIC LOAD

The glycemic index was developed to quantify how rapidly particular foods affect blood-sugar levels as a means to treat diabetes, but it can be used to inform dietary choices for healthy individuals as well.¹⁵ GI is somewhat controversial because it is very difficult to measure, and initially there was no clear standard for how to measure it. As a result, many foods registered wildly different values in early testing. Currently the GI of a particular food is measured by feeding a specified portion to a fasting patient, then drawing blood and measuring blood-sugar levels every 15 to 30 minutes thereafter for up to two hours (see www.glycemicindex.com for more details).⁹ Tests of multiple patients are required to establish the GI of a single food, and even then, the resulting GI measurement can vary sig-



Which will help you lose weight? That depends as the candy has a high GI of 70, but a low GL of 4, while the banana has a low GI of 48, but a high GL of 12. In terms of Glycemic response, the candy is better, if you stop at one. © Fredrik Marmsater

nificantly for that type of food based on the selected portions and ingredients (i.e., how should "pizza" be defined?). The University of Sydney's Human Nutrition Unit has been performing GI testing for years and hosts the International GI Database. GI is assigned to foods on a scale of 0 to 100, relative to pure glucose, which by definition has a GI of 100. A list of GI values for common foods is given on p206-207. The measure of GI shattered some long-held beliefs about which foods are healthy. For example, white potatoes were long thought to be healthy, "complex carbohydrates," but it turns out they have an extremely high GI of 85!¹⁹

Because of the lengthy testing process, only a small percentage of foods have tested GIs, so it is a limited tool for informing dietary choices. Instead of using laboratory-measured values of foods to decide what to eat, it may be more practical to develop dietary rules of thumb.

GI is essentially a measure of how long it takes the sugar in food to be digested and reach the bloodstream. Sugary foods have a high GI because the industrial processing they undergo at the factory accomplishes much of the digestion before they are consumed. The body is capable of digesting *natural* foods, including sugars that are trapped within fruits and vegetables beneath layers of hearty "packaging" (fiber). However, as carbohydrates are further refined in the manufacturing process, the job of digestion has been supplanted by machines. Highly processed carbs thus require very little digestion, and the sugars that were once difficult to access are now instantly available when they hit the stomach. These sugars enter the bloodstream in one big jolt, rather than gradually over a long digestive process. The result is an immedi-

ate blood-sugar spike, along with the aforementioned negative consequences.⁴ This goes for any highly processed carbs — white flour, overly cooked vegetables, corn syrup, etc.

Contrast that with uncooked vegetables or whole grains, both of which require a lengthy digestive process before the sugars contained in them can be accessed. Such foods provide glucose to the blood gradually and thus will not cause the blood-sugar spike that initiates the vicious cycle. Therefore, any foods that are slow to digest into blood glucose, or combinations of foods that slow the breakdown of carb-rich foods, are advantageous. As a rule of thumb, *eat real foods*. Eschew highly processed food products in favor of fresh, uncooked vegetables and fruits, and whole *grain* (not whole wheat) foods. Also, some *combinations* of food help to slow digestion. Fat and protein in general slow the digestion of carbs, as well as acidic foods like vinegar, lemon, and even sourdough bread (that's why balsamic vinaigrette is the salad dressing of choice for healthy eaters).¹ Eating highly processed carbs on an empty stomach is especially harmful in terms of glycemic response, akin to drinking alcohol on an empty stomach. Eat these foods alongside low-GI foods to dampen their effect.

While GI is a helpful metric for evaluating foods, it is incomplete, because it doesn't take into consideration the quantity of food consumed. A piece of hard candy has a high GI (70), but is a small enough portion that it won't spike blood sugar as much as a banana, which has a lower GI (48) but more total carbs. To account for the portion size along with the GI, the glycemic load (GL) is used. GL is defined as:⁹

$$GL = \frac{GI}{100} \times \text{Net Carbs}$$

$$\text{Net Carbs} = \text{Total Carbs} - \text{Dietary Fiber (in grams)}$$

A serving of food with a GL lower than 10 is considered low, and over 20 is considered high. By definition, a 50g serving (a lot!) of pure glucose would have a GL of 50. In the example above, the 6-gram piece of hard candy has a GL of 4, while a 120-gram banana's GL is 12. A favorite snack, corn tortilla chips, has a GI of 63 and GL of 11 for a single 28-gram serving, or about 14 chips (of course, it's hard to eat *only one* serving). For two servings, the GI is still 63, but the GL is 22, etc. That baked white potato mentioned above has 33 net carbs and a massive GL of 26. The authors of the South Beach Diet recommend consuming a running total of fewer than 100 GL points per day for effective weight loss, but their diet is geared toward inactive

folks with heart conditions, not athletes.¹ Instead, use GL as a guide to making healthier choices and learning about food. For those who are right-brained and must count and add, use 100 GL points per day as a baseline, and experiment — climbers in training will likely need more. The application of GL is much more practical than GI when it comes to making dietary choices, and represents a more moderate approach to dieting. Instead of ruling out all sweet and starchy foods, it empowers dieters to consume smaller, more reasonable portions without guilt.

A helpful resource for researching different foods is the *Nutrition Data* website (nutritiondata.self.com),²² which has a large database of foods with nutritional data, including many fast foods and commonly available grocery store items. This interactive resource allows users to input overlooked foods. Because of the difficulty and expense of measuring the GI and GL of foods, they have developed an Estimated Glycemic Load™ or eGL that is a formula for predicting GL based on regression analysis of a given food's known nutrient contents (the formula is compared to lab-tested values on their website, and they demonstrate reasonable agreement). While these estimated values are not perfect, they offer a means of analyzing foods that lack laboratory tested values. They also offer a way to evaluate novel recipes.

Finally, GL is not a panacea. For example, many high-calorie, fattening foods have a low GI/GL (such as a 16-ounce T-bone steak — 1000 calories, 64g fat, GL = 0). Furthermore, recent research indicates that certain *types* of sugar, specifically fructose (and especially high-fructose corn syrup) and sucrose (table sugar), are worse than others. According to researchers, these sugars cannot be absorbed by muscle tissue, so go directly into stored fat.²⁵ Also, beware that GL only measures how *carbohydrates* affect blood-sugar levels, which is only one piece of the dietary puzzle. Understanding and capitalizing on this process skews the caloric equation, but the resulting caloric deficit must be considered as a whole. This is where many fad diets blow it, and why they've been so heavily criticized and often debunked. Absurd amounts of steak and eggs may prevent blood-sugar spikes, but they still make people fat!

Food Table

Nutrition data is shown for some common foods, ranked by GL. Many vegetables have virtually no GL, and aren't listed. GI and GL values are based on clinical trials by the University of Sydney.⁹ Fullness FactorTM, calories per serving, and Estimated Glycemic LoadTM were calculated using the Nutrition Data website and are included to demonstrate how eGL may be used in place of clinical trials.²² Fullness Factor estimates a food's satiety — the higher the number, the more full you'll feel. Pay attention to serving size when comparing foods.

Food Description	Glycemic Index	Fullness Factor TM	Serving Size (g)	Calories Per Serving	eGL TM Per Serving	Glycemic Load Per Serving
Sugars						
Fructose	19	1.4	10	28	4	2
Honey	55	1.4	10	30	5	4
Lactose	46	-	10	-	-	5
Sucrose	68	1.3	10	39	7	7
Glucose	99	-	10	-	-	10
Maltose	105	-	10	-	-	11
Fruits						
Strawberries	40	4.3	120	40	2	1
Cherries	22	3.4	120	52	4	3
Grapefruit	25	4.5	120	32	3	3
Plums	24	3.5	120	52	4	3
Apples, Braeburn	32	3.3	120	65	3	4
Pears	33	3.1	120	64	4	4
Watermelon	72	4.5	120	32	4	4
Oranges	48	3.5	120	52	4	5
Pineapple	59	3.3	120	64	4	7
Grapes	46	2.8	120	76	8	8
Bananas	48	2.5	120	105	10	12
Raisins	64	1.6	60	168	26	28
Vegetables						
Carrots	47	3.8	80	32	3	3
Pinto Beans	14	2.7	150	214	13	4
Black Beans	30	2.8	150	198	12	7
Sweet Corn	60	2.4	80	83	8	11
Yams	37	2.2	150	114	16	13
Sweet Potato	61	2.5	150	138	14	17
Instant Mashed Potato	97	2.2	150	169	9	19
Baked Potato	85	2.5	150	145	16	26
Breads						
Barley flour bread, whole meal	43	-	30	-	-	5
Rye, Whole-grain, pumpernickel	46	-	30	-	-	5
Sourdough Rye Bread	53	2.0	30	83	8	6
Healthy Choice 7 Grain Bread	55	-	30	-	-	8
Wheat Tortilla	30	1.8	50	146	15	8
Old El Paso Corn Taco Shells	68	1.5	20	98	8	8
Healthy Choice 100% Whole Grain Bread	62	2.2	30	74	6	9
White Bread	70	1.9	30	101	14	10
English Muffin	77	2.0	30	64	8	11
Wheat Bread	58	2.0	30	74	8	12
Corn Tortilla	52	-	50	-	-	12
French Baguette, White	95	2.0	30	92	11	15
Bagel, White	72	2.0	70	190	22	25

Food Description	Glycemic Index	Fullness Factor TM	Serving Size (g)	Calories Per Serving	eGL TM Per Serving	Glycemic Load Per Serving
Breakfast Foods						
Yogurt	36	3.2	200	126	8	3
Oatmeal, Plain	25	2.1	30	106	11	5
Oatmeal, Apple & Cinnamon	37	1.9	30	128	15	8
Kellogg's All Bran	38	3.2	30	81	8	9
Kellogg's Raisin Bran	61	2.1	30	90	12	12
Grape Nuts	75	2.1	30	101	14	16
Kellogg's Corn Flakes	92	1.6	30	101	18	24
Mixed Meals						
Pizza Hut Super Supreme, Thin Crust	30	2.1	100	209	10	7
Bean Burrito with Tomato Sauce	39	2.3	100	211	12	9
Pizza Hut Vegetarian Supreme, Thin Crust	49	2.1	100	175	11	12
Spaghetti, Whole Meal	32	-	180	-	-	14
Pizza, Cheese, Thick Crust	60	2.0	100	272	17	16
Spaghetti, White	42	2.1	180	284	29	20
Couscous	61	2.3	150	176	18	21
Peanut Butter Sandwich	59	-	100	-	-	26
Snacks						
Peanuts	14	1.5	50	297	1	1
Ice Cream, Low-Fat (1.2% fat)	47	2.2	50	69	7	5
Peanut M & M's	33	1.5	30	144	10	6
Chocolate Pudding	47	2.2	100	119	9	7
Ice Cream, Regular	61	1.8	50	103	6	8
Popcorn	72	2.2	20	78	8	8
Ironman PR Bar	39	-	65	-	-	10
Potato Chips, Plain	51	0.9	50	280	13	12
Pretzels	83	1.8	30	113	16	16
Rice Cakes	78	1.7	25	108	15	17
Tortilla Chips	63	1.5	50	246	20	17
Twix Bar	44	1.3	60	291	24	17
Chocolate Cake, Betty Crocker	38	1.5	111	367	30	20
Snickers Bar	68	1.5	60	271	21	23
Pop Tarts, Double Chocolate	70	1.5	50	201	25	24
Power Bar	56	2.0	65	247	29	24
Skittles	70	1.2	50	202	32	32
Beverages						
Tomato Juice	38	-	250 mL	-	-	4
Skim Milk	32	4.2	250 mL	86	9	4
Soy Milk, Low Fat (1.5% fat)	44	-	250 mL	-	-	8
Chocolate Milk	34	3.1	250 mL	157	13	9
Apple Juice	40	3.5	250 mL	114	6	12
Gatorade	78	4.5	250 mL	63	1	12
Orange Juice	57	3.6	250 mL	110	9	15
Coca Cola	63	3.7	250 mL	117	5	16
Cranberry Juice	68	3.2	250 mL	137	8	24

WHAT TO EAT

This dietary trifecta of metabolism, appetite, and fat storage are all affected by the body's response to carbohydrates. A perusal of nutrition labels will quickly reveal that carbs pack a lot of calories into popular foods. While fat has the highest caloric content by weight (nine calories per gram of fat vs. four per gram for carbs or protein), the fact is, modern foods have significantly more carbs than fat, so the majority of calories consumed come from carbs. This is by design. To meet the demand for low-fat foods, manufacturers replace the small portions of fat with loads of processed carbs, generally *increasing* calories and disrupting the glycemic response.

It should by now be obvious why commercial diets shun carbs. However, the real key is to eliminate highly processed carbs (refined sugar, refined flour, corn syrup, etc.). Naturally occurring carbs such as those locked within vegetables, fruit, and whole grains are much less harmful than refined sugars. It is not necessary to completely eliminate food groups. Such approaches are impractical, and typically lead to failure. Instead, eat a little of everything, and a lot of the right things.

VEGETABLES, FRUIT, AND WHOLE GRAINS

A healthy, high-performance diet should consist of large portions of fresh vegetables — the rawer, the better. Salad is perfect, and can be eaten in nearly unlimited quantities while dieting, as long as it's not

What I Eat When I'm on a Diet

Taste is personal, so take this list with a grain of salt....

- *Always:* Water...I never drink calories when I'm dieting.
- *Mostly:* Salad — tons of it; two to three salads *per day*, with lots of fresh veggies and olive oil-based balsamic vinaigrette dressing. I often top it with light tuna fish or grilled chicken. Salad comes in many varieties. If the typical garden salad sounds boring, taco or mango salads are just the tip of the iceberg; there are countless tasty salads to try.
- *Occasionally:* Low GI fruit — grapefruit, cantaloupe, apples, strawberries, and when I want to feel "full," watermelon!
- When I want *meat*: Grilled and seasoned fish and chicken breast (with fat removed).
- When I want *bread*: Whole grain breads (rye) and crackers (Wasa® Light Rye) — topped with tuna salad, salsa, guacamole, hummus, or bruschetta for protein and flavor.
- When I want *crunchy*: Quaker® Multigrain Fiber Crisps and Popped Rice Snacks (miniature rice cakes in both sweet and savory flavors), and pickles, carrots, celery sticks, etc.
- When I want *dessert*: Mango or peach salsa on a rice cake or Crisp 'N Light Wasa® cracker, pumpkin pie sans crust, dark chocolate-dipped strawberries, or Breyers® low-sugar ice cream.
- When I *road-trip*: Subway® grilled chicken salad with all the veggies; no cheese, croutons, or creamy salad dressings.
- When I want to *prevent mindless snacking*: Sugar-free chewing gum and Tic-Tacs®.
- When I *train or climb*: Trader Joe's "Power Crunch" protein bar, Clif Builder's Bar, Fiber One® granola bars, and/or Kellogg's® Fiber Plus granola bars.

— Mike Anderson



An assortment of Mike's go-to diet foods: plenty of greens, lean protein, and fiber.

Mike Anderson

"I went on a strict diet for about a week. Salads only....This wasn't hard. I was keen to lose the weight to give me an edge on that first move and I was so excited about doing [The Dominator, V13] that I could hardly eat any-way." — Jerry Moffatt¹⁸

smothered in bacon bits, croutons, ham, and fatty ranch dressing — again, raw vegetables should dominate. In general, fruits have more sugar and therefore calories than vegetables, but they provide numerous vitamins and minerals. Dieters can eat *almost* as much fruit as they want, but realize it can add up. Some fruits are better than others; refer to the GL data in the table on pp.206-207 (Grapefruit, cantaloupe, and berries are better than bananas, tomatoes, and watermelon.)

Besides slowing digestion and glycemic response, another great benefit of fruits and vegetables is that dieters can eat enough of them to feel full without ingesting excessive calories. This is because they contain mostly water and dietary fiber that isn't absorbed by the body. Researchers at the University of Sydney, who have been responsible for the lion's share of GI and GL research, and at Penn State University (responsible for the popular *Volumetrics* diet) have sought to identify satiating food properties that stave off hunger.^{13, 23} *Nutrition Data* (nutritiondata.self.com) has extended that idea in an effort to identify foods that are the most filling per calorie. Their *Fullness Factor*™ is an analytically derived satiety metric calculated based on the nutritional contents of a given food, and correlates well with lab-tested data.²² Use this resource as a guide to learn about foods and identify diet-friendly choices. Not surprisingly, bulky fruits and vegetables top their list of low-calorie, filling foods.

Finally, foods high in processed carbs, such as most bread and crackers, should be limited. Opt for foods that are made from whole grains (not whole wheat). Rye bread, Wasa® whole grain rye crackers, Ezekiel bread, steel-cut oats, and whole grain pastas, to name a few, are good options. Highly refined breads, breakfast cereals, oatmeal, chips, and crackers have the detrimental combination of high GI and high net carbs. These should be eaten sparingly, and preferably in conjunction with low-GL foods like vegetables, protein, or fat to slow the glycemic response.¹

LEAN MEAT

With a plate mostly full of veggies, the remaining space should contain lean protein: fish or chicken. Protein is essential for everyone, but especially athletes. The body relies on it for numerous functions. In the context of weight management, the challenge is always to get sufficient protein without excessive calories; enter lean meats. These will ideally pack a lot of protein into few calories. Tuna fish (canned, in water) may be the ultimate climber protein source, being cheap, lean, and non-perishable. Bodybuilders eat so much tuna that they can often be picked out of a dark room by their "tuna breath." A three-ounce serving has only 100 calories, almost all of which come from a whopping 22 grams of protein! For those with more refined tastes (and income), fresh fish and chicken breast are generally better tasting, while also good sources of protein. Fish has beneficial omega-3 oils, but can be expensive. In case it's not obvious, these meats should be served grilled or baked, not deep-fried.

Pork is not a lean protein source and should be avoided when dieting. Soy-based protein bars are a convenient protein fix, but they will generally pack in more calories, with lots of sugar added to improve taste as well as fat. Clif's protein bar requires nearly 300 calories to get the same amount of protein as a 100-calorie serving of tuna. That's because these bars are also designed to replenish carbs and fat after a hard workout. Soy protein in general is very controversial, and government regulatory bodies have recently stripped some soy products of previous health claims.⁶ Those who eat a lot of soy products are encouraged to research its safety to form their own opinion. Beans are a good source of protein, but some beans (black and refried) are packed with fat and carbs as well.

FATS

Fats are necessary, and should be consumed in moderation to maintain strong tendons, ligaments, and cartilage. When eaten together with high-GI carbs, small portions of fat can slow the digestion of these carbs (as explained above) and they can be satiating, making it easier to stick to the diet. Foods naturally have a combination of saturated, monounsaturated, and polyunsaturated fats. The healthiest foods have the highest proportion of poly- and monounsaturated fat and the least proportion of saturated fat. Monounsaturated fats are known to lower LDL (bad) cholesterol, thus benefiting heart health, and they have an anti-inflammatory effect — of great benefit to climbers.^{1, 19}

Common food sources of monounsaturated fat that are also low in saturated fat are safflower oil, olive oil, canola oils, avocado, and some nuts (macadamia, almonds, cashews, and pecans).²⁶

Omega-3 fatty acids, a form of polyunsaturated fat,²⁷ are also known to reduce inflammation. Omega-3s are mostly found in fish and other seafood, as well as flaxseed oil, kiwifruit, and certain supplements.¹⁶ Omega-6 fatty acids are also a type of polyunsaturated fat that may be *detrimental* to climbers, because they are known to *increase* inflammation and negate the benefits of omega-3s. Government bodies recommend consuming a 1+:4 ratio of omega-3s to omega-6s (eat one gram or more of omega-3s for every four grams of omega 6s).²⁴ Most vegetable oils have low ratios of omega-3 to omega-6, but the worst are peanut oil and sunflower oil (no omega-3), corn oil (1:46), and palm oil (1:25). All of these should be limited or

avoided, if possible.²⁰ Foods with high proportions of omega-6 include eggs (1:15), avocado (1:15), and most nuts (ratios vary).²² Except for fish-gobbling grizzly bears, it is nearly impossible to avoid high ratios of omega-6s in any diet. Therefore, omega-3 fish-oil supplements are highly recommended to boost omega-3s, thus improving the overall ratio.

As usual, the best fat sources are those that are least processed. Olive oil has numerous health benefits, and should be the first choice for cooking and dressing foods (especially salad and bread). Extra-virgin olive oil is the healthiest (and most expensive), being highest in monounsaturated fat and with the highest omega-3 to omega-6 ratio, and has been shown to displace the negative effects of omega-6s.^{11, 24} Olive oil is also known to lower blood sugar, so it can be used to dampen the impact of other, high-GL foods.⁸

Saturated fat and trans fat should be reduced or avoided. Saturated fats are mostly found in animal-based products, but also in some vegetable oils (common sources are fatty red meats, cream, cheese, butter, lard, coconut oil, palm seed oil, and chocolate). Saturated fat raises cholesterol, and the American Heart Association recommends restricting intake to less than 7 percent of total calories (around 15g per day).¹⁷

Trans fat occurs naturally, but in the modern diet it comes primarily from hydrogenated oils (margarine) — another “miracle” product invented in a food lab. Trans fat raises LDL cholesterol (bad) and lowers HDL (good), delivering a double-whammy to the heart. Trans fat is common in fast food, snack foods, fried foods, and highly processed baked goods. It is used because it is cheap, increases shelf life, and decreases refrigeration needs — all benefits for the food industry, but not the consumer. Experts recommend eliminating all trans fat, and some government bodies have outlawed it.⁷ The best way to know if a particular snack has trans fat is to parse the label for ingredients containing the word “hydrogenated.”



Dietary choices can affect the body's ability to fend off inflammation. For example, Omega 3 oils improve joint health and may help climbers recover from tweeky moves. Chris Hirsch chugging through mono country on *Big Train*, 5.13a, Spearfish Canyon, SD.  Andrew Burr

Reducing Inflammation

Some foods have been shown to reduce inflammation, which is a must for rock climbers in training. Here are some dietary ways to reduce inflammation:

- Eat monounsaturated fats — safflower, olive and canola oils, avocado, and some nuts
- Eat omega-3 fatty acids — fish (such as herring, salmon, mackerel, sardines, and tuna), algae, and fish-oil supplements
- Avoid omega-6 fatty acids — found in vegetable oils such as peanut, sunflower, corn, and palm oil
- Eat certain spices — ginger, curry powder, garlic, onions
- Maintain stable blood sugar (high blood sugar aggravates inflammation)

DAIRY PRODUCTS

Dairy products are somewhat controversial, and whether or not to include them in a healthy diet is a matter of taste and personal choice. In moderation, dairy products aren't overly harmful, and they can be a good source of protein. Cheese may be the worst form of dairy (with lots of saturated fat and calories), and the easiest to reduce or eliminate. The average American eats 33 pounds of cheese a year, three times the level of the 1970s. This meteoric rise in cheese consumption is due in part to a concerted effort by "Dairy Management" (a partnership between the dairy industry and the USDA) to get Americans to eat more cheese, in order to help dairy farmers make more money. Therefore, whatever subconscious impulse is compelling people to include lots of dairy in their diet was likely surreptitiously placed there by the false advertising machine of a real-life conspiracy. These "public servants" are responsible for such gems as stuffed-crust pizza, Burger King's Cheesy Angus Bacon Cheeseburger, and the "Got Milk?" ad campaign. If pizzas or burritos are a must, ask the kitchen for "half" or "light" cheese.

Skim chocolate milk has been promoted as the ultimate post-workout recovery drink because of its mix of lean protein and medium glycemic index. It is clearly superior to protein-free beverages like Gatorade, but a protein bar eaten with plenty of water will also do the trick.

SNACKS AND TREATS

Clearly, snacks and treats are not diet foods, but realistically, people are going to cheat from time to time. The trick is to cheat in the best possible way. There are a number of healthier snacks that are sweet enough to provide the necessary fix without initiating hypoglycemic shock and derailing the diet.

Generally, it is best to eat sweets along with foods that slow the glycemic response (fiber, fat, or acid). Fruit is a good starting point, but also consider fruit salsas, or berries dipped in dark chocolate. Fatty deserts like pudding, frozen yogurt, ice cream, and pumpkin pie have relatively lower GIs so they are better choices than cookies, cakes, and candy bars. The first couple bites are the best, so keep the portions small.

For salty and savory snacks, avoid fried chips and crackers, which have lots of saturated fat and calories. Light popcorn with little or no butter can provide similar satisfaction, but don't eat an entire bag in one sitting — a little goes a long way. Also, Quaker makes miniature rice cake snacks in several flavors that provide the sensation of eating flavored corn or potato chips, but without the saturated fat and calories. Cut vegetables like carrots, cucumbers, cauliflower, and celery sticks sprinkled with salt and other seasoning are the best choice for a savory fix. Pickles have almost no calories, and make a nice quick snack.



If you have a sweet tooth, select sweets such as these that will scratch the itch without wrecking your diet.  Mike Anderson

Dealing with a Sweet Tooth

I have a vicious sweet tooth, so I struggle with excluding sweet treats when it's dieting time. Nevertheless, I've found that if I go cold turkey for two weeks, my cravings are markedly reduced, and I can be satisfied with less-sweet foods than I would otherwise want. This conditioning is well-known; we become desensitized to flavors over time, necessitating richer and richer foods in order to get the same satisfaction. Therefore, as you eliminate the most egregious refined sweets from your diet, you should be able to satisfy the cravings with real, less-processed sweet foods such as fruit. In fact, after many weeks of omitting refined sugar, I find rich deserts like chocolate cake to be downright unpalatable. If you're also a slave to desert, have a small piece of fruit after dinner instead — before long you won't miss the other stuff.

— Mike Anderson

VITAMINS

When dieting, it is especially important to take vitamins and omega-3 fish-oil supplements, because it is unlikely that they will all be obtained naturally in 1,500 to 1,800 calories of food per day. Nutrient content varies by brand names and growing locations (in the case of fruit and vegetables), so it is very difficult to definitively know what is and isn't included in a given diet. It is much easier to take a multivitamin supplement and be certain. In addition to a multivitamin, climbers should consider taking a *glucosamine with chondroitin* supplement, which contains the building-block materials needed to repair joint cartilage. While the effectiveness of this supplement has not been proven, many orthopedic surgeons and injury-prone athletes swear by it. In the same vein, an omega-3 supplement is a must for its cardiovascular and anti-inflammatory benefits.

Super Subs

For most people, the hardest part about dieting is giving up your favorite foods or snacks. However, if you are willing to experiment, you can often find substitutes for your favorite foods that are much healthier, but provide similar satisfaction. For example, my Achilles' heel is chips and salsa. Salsa is actually a very healthy food (except for the sodium content), but tortilla chips are awful. A paltry 14-chip standard serving has 140 calories, and an eGL of 11. The problem is, I can't stop at 14 chips, and it's easy for me to consume an entire jar of salsa and half a bag of chips in one sitting. Frankly, the main purpose of the chips is to deliver salsa to my mouth. Recently, I discovered that whole grain Wasa® crackers can do that just as easily, and with a much lower GL and better feeling of satiety (a spoon works too). Some other good substitute foods include:

- Low-sugar ice cream over high-sugar ice cream
- Dark chocolate over milk chocolate
- Apple over banana
- Oatmeal-raisin cookie over chocolate-chip cookie
- Snickers marathon bar over [insert your favorite high-sugar energy bar]
- A taco salad or burrito bowl over tacos or burritos
- Fish/chicken/turkey over red meat
- Rice cakes over chips/crackers



Mike Anderson

An easy way to shave calories is to make a few substitutions. Identify the worst GI/GL offenders in your current diet and find a tasty alternative.

Sample Daily Meal Plan

This is a potential baseline meal plan for weight loss. It's only 1600 calories (appropriate for a 140-pound climber), but provides regular snacks throughout the day. Many items can be easily substituted to account for taste (i.e., chicken vs. fish, protein bar for breakfast, etc.). If weight drops too quickly, add calories with simple substitutions, such as a granola bar instead of an apple.²²

Food	Portion	Cal (%Total)	Fat	Protein	Carbs	FF	eGL
BREAKFAST (9:00)		229 (15%)	11 g	13 g	19 g	-	6
Hard-boiled Egg	Qty: 2.0 x 1 large (50g)	155 (10%)	11 g	13 g	1 g	2.5	2
Grapefruit	Qty: 2.0 x 1/2 fruit (3-3/4" dia) (123g)	74 (5%)	0 g	1 g	18 g	4.5	4
SNACK 1 (10:30)		95 (6%)	0 g	0 g	25 g	-	5
Apple	Qty: 1.0 x 1 medium (3" dia) (182g)	95 (6%)	0 g	0 g	25 g	3.3	5
SNACK 2 (11:30)		49 (2%)	0 g	3 g	10 g	-	3
Carrots	Qty: 8.0 x 1 strip large (3" long) (7g)	23 (1%)	0 g	1 g	5 g	3.8	2
Cauliflower	Qty: 2.0 x 1 floweret (13g)	6 (0%)	0 g	1 g	1 g	4.5	0
Celery	Qty: 8.0 x 1 strip (4" long) (4g)	5 (0%)	0 g	0 g	1 g	4.5	0
Broccoli	Qty: 1.0 x 1/2 cup, chopped (44g)	15 (1%)	0 g	1 g	3 g	4.5	1
LUNCH - Mixed Green Salad with Tuna Fish (1:00)		232 (14%)	10 g	24 g	13 g	-	2
Tuna Fish, light, in water	Qty: 1.0 x 3 oz (85g)	99 (6%)	1 g	22 g	0 g	3.4	0
Romaine Lettuce	Qty: 3.0 x 1 leaf outer (28g)	15 (1%)	0 g	1 g	3 g	4.5	0
Cucumber	Qty: 0.5 x 1/2 cup slices (52g)	4 (0%)	0 g	0 g	1 g	4.5	1
Green Bell Peppers	Qty: 1.0 x 10 strips (27g)	5 (0.5%)	0 g	0 g	2 g	4.5	0
Onions	Qty: 0.5 x 1 slice, 1/4" thick (38g)	8 (0.5%)	0 g	0 g	2 g	3.8	0
Tomatoes	Qty: 2.0 x 1/4 wedge (31g)	11 (1%)	0 g	1 g	2 g	4.5	1
Dressing, Bals. Vinaigrette	Qty: 1.0 x 2 Table Spoons (30g)	90 (5%)	9 g	0 g	3 g	1.2	0
SNACK 3 (3:00)		123 (8%)	1 g	1 g	32 g	-	10
Grapes	Qty: 2.0 x 1 cup (92g)	123 (8%)	1 g	1 g	32 g	2.8	10
SNACK 4 (4:30)		192 (11%)	0 g	7 g	42 g	-	17
Light Rye Wasa Crackers	Qty: 2.0 x Custom Food (17g)	120 (7%)	0 g	4 g	28 g	3.0	14
Salsa	Qty: 2.0 x 100 grams (100g)	72 (4%)	0 g	3 g	14 g	4.1	3
DINNER - Baked Fish with Salad & Buttered Bread (6:30)		513 (30%)	17 g	55 g	32 g	-	12
Baked Grouper (Fish)	Qty: 1.0 x 1 fillet (202g)	238 (15%)	3 g	50 g	0 g	3.3	0
Romaine Lettuce	Qty: 2.0 x 1 leaf outer (28g)	10 (1%)	0 g	1 g	2 g	4.5	0
Spinach	Qty: 1.0 x 1 cup (30g)	7 (0%)	0 g	1 g	1 g	4.5	0
Cucumber	Qty: 0.5 x 1/2 cup slices (52g)	4 (0%)	0 g	0 g	1 g	4.5	1
Green Bell Peppers	Qty: 1.0 x 10 strips (27g)	5 (0.5%)	0 g	0 g	2 g	4.5	0
Onions	Qty: 0.5 x 1 slice, 1/4" thick (38g)	8 (0.5%)	0 g	0 g	2 g	3.8	0
Banana Pepper	Qty: 1.0 x 1 ounce (28g)	8 (0%)	0 g	0 g	1 g	4.5	1
Dill Pickles	Qty: 1.0 x 1 spear (35g)	4 (0%)	0 g	0 g	1 g	4.5	0
Dressing, Bals. Vinaigrette	Qty: 1.0 x Custom Food (30g)	90 (5%)	9 g	0 g	3 g	1.2	0
Bread, rye	Qty: 2.0 x 1 slice, thin (20g)	103 (6%)	1 g	3 g	19 g	2.0	10
Butter	Qty: 1.0 x 1 pat (1" x 1/3") (5g)	36 (2%)	4 g	0 g	0 g	0.5	0
Snack 5 - Desert (8:00)		172 (11%)	1 g	3 g	43 g	-	6
Watermelon	Qty: 2.0 x Slice, ~ 1/16 melon (286g)	172 (11%)	1 g	3 g	43 g	4.5	6
TOTAL		1605	50 g	106 g	216 g	-	73

BASELINE DIET

It should by now be clear that slow-digesting, low-GI foods help control appetite, keep metabolism high throughout the day, and discourage fat storage. However, that is only part of the equation. In order to lose weight, the burden remains to take in fewer calories than are burned. Some diets ensure that this is accomplished by counting calories or tracking points. This can be tedious at best, often impossible, and nearly always short-lived. A more common-sense approach is to begin with a baseline diet that can be replicated every day. A dieter can eat this consistently for a week, track his weight, and evaluate if it's working. For example, a typical daily meal plan is shown on the previous page. If the weight loss is too slow, gradually

reduce certain portions (such as the amount of salad dressing at lunch, the daily allotment of fruit, or the serving of meat for dinner) until the desired weight loss occurs. If the weight loss is too rapid, increase portions slightly. After a few weeks of experimentation, it will be evident what the right diet plan is to achieve the desired rate of weight loss. Once determined, the baseline diet can be used season after season to lose weight.

CARDIO CONUNDRUM

Increasing exercise volume to boost weight loss (and avoid sacrificing food) is a common strategy that can be detrimental to climbers. First, the Rock Prodigy program provides ample exercise already, so additional exercise will hamper recovery and interfere with climbing-specific training. Second, it can be easier to lose weight by *reducing* the volume of cardiovascular exercise. Long, moderate-intensity cardio substantially increases food cravings, and often adds lower-body mass that isn't needed for climbing. This is a classic blunder, on par with getting involved in a land war in Asia.

Many dieters feel they *deserve* a good meal after a hard workout, but they often *deserve* too much, completely wiping out the weight-loss benefits of the workout. For example, it takes one 270-calorie donut to wipe out the benefits of a three-mile hike. Therefore, climbers opting for aerobic exercise must avoid overeating after hard workouts. Eat the minimum needed for recovery (half a protein bar or glass of chocolate milk) and drink plenty of water (at least 16 ounces), and then move on to another distracting activity. Most climbing workouts described here target the forearms, a very tiny muscle group with minimal glycogen requirements. A massive post-workout binge is not necessary to ensure adequate recovery.

Keep in mind, most diet and fitness plans aren't intended to cause *weight* loss; rather, they are intended to cause *fat* loss by increasing muscle mass and decreasing adipose fat. In other words, these plans may actually *increase* weight. This is fine for most people, but not climbers. Climbers need to be lightweight — leanness is just a side effect. Furthermore, pop diets are intended to change lifestyles in order to lose weight gradually over time and keep it off, without yo-yo-ing. This is nearly the opposite of what a peaking climber desires, which is to quickly lose 10 pounds or so for the Performance Phase before returning to the ideal body composition for training a few months later.

The Road-Tripping Dilemma

Maintaining a diet while on the road is hard for anyone, and climbers tend to be on the road a lot. Unfortunately, easily transportable, nonperishable foods are almost always unhealthy (high in sugar and trans fat); while the restaurant menu options are loaded with extra calories and unhealthy ingredients. Here are some simple tips to purify your next road trip:

 Bring a cooler and pre-pack it with healthier snacks like baby carrots and other veggies, apple slices, watermelon slices, strawberries, etc.

 For a very long trip, go online and scope out the locations of grocery stores near the highway so you can restock any perishable, healthy snacks en route. Also, search for healthier casual-dining restaurants like Quizno's®, Qdoba®, Panera Bread®, etc. A little pre-planning like this may prevent you from caving in to the Golden Arches.

 When fast food can't be avoided, look for restaurants that provide choices (and nutrition data) like Subway®, which are co-located in many gas stations. Just be careful — it's very possible to overeat at a healthy restaurant. Stick to the salads and forgo the fatty meats, cheese, heavy dressings, sodas, chips, and cookies.

 In the car, set time goals for eating. For example, have a small snack no more than every hour or 90 minutes. Don't leave that bag of pretzels propped open within arm's reach; pack it up tightly or put it in the trunk after you've dished out the right portion.



Despite Lena Moinova's beta for gaining the start holds on *Toker* at the Red River Gorge, KY, it's a rare climb indeed that demands much leg strength, therefore, rock climbers generally don't need large, muscular legs.  Mike Anderson

GENERAL STRATEGIES & TACTICS FOR CONTROLLING DIET

In addition to knowing *what to eat*, successful dieting requires habitual lifestyle changes. These are proven methods for successful dieting:

ENVIRONMENTAL ADJUSTMENTS TO PROMOTE WEIGHT LOSS

- **Track progress:** Get a good bathroom scale and weigh in at the same time each day (weight fluctuates throughout the day). Tangible weight loss is highly motivating. The experts recommend losing no more than two pounds per week, but those recommendations are for people who want to keep the weight off permanently and are accounting for converting fat to muscle. Climbers are merely trying to quickly drop down for four to six weeks before returning to a higher training weight, so up to four pounds a week is reasonable.

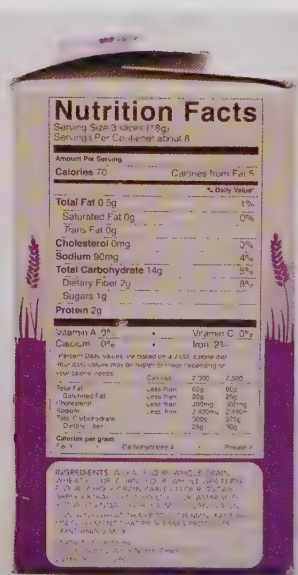
- **Keep food out of sight and hard to access:** Tuck food safely away behind a cupboard door or in a desk drawer, and don't leave tempting snacks around the house. Every time you see food, you are tempted. Better to keep it out of sight, out of mind. One study recorded that folks ate 70 percent more out of a transparent candy dish than a solid-opaque one.²⁹ If you have family or roommates who aren't dieting, store your food in a separate cupboard and keep out of the others' so you won't be tempted by their treats. Make whatever food is around hard to get to — in one clever experiment, secretaries ate more than twice as many chocolates throughout the day when a bowl was placed on their desk rather than across the room.²⁹
- **Motivate the right choices:** Hang a picture of your goal route or other motivational materials at the "gates of temptation," be it the fridge door, pantry, or on that box of cookies. At your moment of weakness, this will remind you of the big picture.
- **Dress skinny:** Wear your tight-fitting, "in shape" clothes to remind you of your goal, and discourage yourself from overstuffing at mealtime.

BEHAVIORAL ADJUSTMENTS TO PROMOTE WEIGHT LOSS

- **Graze:** Instead of eating two or three massive meals a day with forced fasting in between, try grazing, or eating numerous small meals and snacks throughout the day. Aim to eat something every one and a half to two hours, be it a piece of fruit, bowl of vegetables, or a small-portion meal.
- **Distract yourself:** Engage in activities that take your mind off food or make it difficult to eat altogether. Climbing is the ultimate distraction, because you are nearly always climbing or belaying, and eating becomes an inconvenience. When that isn't practical, engross yourself in your work or a personal project. Even playing a demanding video game can help; it's hard to spoon ice cream while pushing "A-A-B-Square."

"How hard it is to lose weight when you're overweight! That's where the mental process is interesting, because you face your choices: Do I really want to do harder routes or do I content myself with my cushy routine? At what point do I make the effort to climb harder?" — J.B. Tribout³¹

- **Use your head:** The brain needs as much as 20 percent of your daily caloric intake. Activities that keep you thinking hard will speed up your metabolism, and may be why mindless activities like watching TV are so closely linked with obesity.
- **Read labels and learn about food:** You can't expect to lose weight while remaining ignorant about nutrition. Spend some time in the grocery store reading labels, and do the shopping yourself. All the junk food is usually in the middle aisles, so stick to the perimeter of the store where the fresh food is. Healthy food takes more work to prepare than convenient junk, but it's worth the effort. Don't bring foods into the house that don't support your goals, and learn how to cook healthy, tasty meals.
- **Work as a team:** If you can, surround yourself with like-minded folks. Studies have shown a direct correlation between the number of people at a meal and how much is consumed, which is to say, if other people are eating a lot, you will too. On the other hand, if you are surrounded by supporters with similar goals, you can encourage each other to stick to the plan.
- **Be careful eating out:** Restaurants are traps, designed by cunning geniuses to encourage overeating. Also, the food is usually very rich, with way too much butter, lard, and sugar. Ask the waiter to skip the bread bowl or take it back early. Put half your meal in a to-go box or split an entrée with someone else. If dessert is unavoidable, share it with the whole table.



Learn about the foods you eat, and use that information to make smart choices at the grocery store.  Mike Anderson

TIPS FOR CONTROLLING PORTIONS

Sometimes the hardest part of sticking to a diet isn't the fasting between snacks and meals, but controlling the portions eaten during meals. People often use illogical, external cues to tell them when they are full, rather than listening to their own bodies, as documented by Dr. Brian Wansink, author of *Mindless Eating: Why We Eat More Than We Think*. As a behavioral psychologist and professor, he has performed numerous experiments to explore these phenomena.²⁹ The following techniques will help control portion sizes:

- **Plate food in the kitchen:** Serve food onto plates in the kitchen, then retire to the dining room to eat. If you eat next to the leftovers, you'll likely eat more.
- **Use smaller dishes:** Numerous studies show that, given the exact same portion size, we feel more satiated and consume less when food is served in a smaller dish that creates the illusion of a full plate.²⁹
- **Eat voluminously:** The authors of *Volumetrics* have clinically proven that people judge satiety by the visual size and weight of food, not calorie content or cues from the stomach.²³ Therefore, add volumizing food to every meal — high-bulk fruit and vegetables, especially salad. It creates the look of a full plate that subconsciously triggers satiety.
- **Seal containers:** To prevent repeated snacking, dish out the food and then put away the bag — tape it shut and place it in a hard-to-reach spot...or the garage or basement.
- **Repackage big-box food:** The "Grande" bag of tortilla chips is more economical, but studies show that bigger bags encourage overeating every time. The moment it comes home, separate the contents into smaller Ziploc bags of more appropriate portions. When it's time for chips and salsa, even if you blow through multiple Baggies, you will eat much fewer chips than if you were eating from the "Grande" trough.
- **Slow down:** Eating quickly increases the chances you will overeat. Put your fork down after every bite, chew and swallow before taking the next bite, and if eating with others, try to be the last one to finish.
- **Reduce variety:** Familiarity breeds contempt, which can be used to your advantage. A wide variety of foods will compel you to eat more of each type. If your dinner plate only has two courses on it, you're likely to eat less than if you had four.²⁹

- *Don't extend meals:* After dutifully controlling portions during a meal, don't waste the effort by returning to the kitchen to snack. Before the meal starts, plan an activity to immediately follow it — a casual bike ride, a movie, etc. Make it a time crunch — you won't linger in the kitchen if you're running late for *Star Wars 8: The Return of Interesting Dialogue*. Another technique is to eat a breath mint, chew a stick of gum, or brush your teeth as soon as the meal is over (while you are still "full").
- *Retain some comfort foods:* Eliminating *all* of your favorite foods is very difficult and often leads to failure. While many comfort foods are health nightmares (like pizza and fried chicken), most people have comfort foods, *or versions of them*, that are perfectly reasonable. Watermelon, salsa, or even a thin-crust vegetarian pizza with light cheese can satisfy a comfort-food craving without the guilt.
- *Don't be fooled by "healthy" advertising:* Labels such as "low-fat," "low-carb," "all-natural," etc. are misleading. For example, so-called "low-fat" granola has only 10 percent fewer calories than regular granola because the fat's been replaced with sugar. Pay attention to the label on the back of the box, not the one on the front, and always take note of the suggested serving size — for example, a 20-ounce soda contains 2.5 servings.
- *Create food policies:*²⁹ Many times, it is easier to rule out certain food options than it is to allow them in moderation. For example: no cheese, no beef, no liquid calories, no exceptions. There is no ambiguity, and it's an easy policy to enforce. Another example of an easy-to-follow policy might be to eat only "on the hour." Such policies will become habits after a few weeks, and habits are hard to break.

DIETARY CHOICES TO PROMOTE WEIGHT LOSS

- *Don't drink calories:* Sugary drinks, exotic coffees, and alcohol all pack in calories with little satiating benefit. Drink plenty of water, and spare those calories for foods that provide essential nutrients and satisfy hunger. If you must drink, studies show that using a tall, skinny glass will result in drinking 20 percent less than if using a short, wide glass.²⁹
- *With treats, a little goes a long way:* The first few bites of a sweet treat or savory snack are the most satisfying, and then the law of diminishing returns kicks in. You'll feel just as satisfied after eating two cookies as you will after eating a dozen, with the minor difference being the self-loathing associated with the latter case.

SUMMARY

- Climbers should strive for general fitness, but only need to be extremely lean during their Performance Phase.
- Excessive calorie restriction is counterproductive during the Strength Phase.
- A weight-loss diet doesn't need to exclude entire food groups (i.e., carbs). Numerous studies show dieters are more likely to stick to a diet that is inclusive.
- All things in moderation: Snacks may be eaten occasionally, but don't eat the whole piece of cake.
- Eat plenty of salad and raw veggies, and fruit within reason — they are filling and aid digestion.
- Regardless of the diet strategy, carbs must be limited in order to keep calories down. GL is a good way to evaluate foods, track calorie intake, and make dieting less difficult.
- Olive oil, fish, fish oils, and some nuts are good sources of satiating fat.
- Eat plenty of lean protein, preferably fish and chicken.
- Be careful not to overeat after workouts.
- Supplement the diet with a multivitamin, glucosamine with chondroitin, and omega-3s.
- Stay busy, don't think about food, dish out reasonable portions, and stop at the right time.
- Be mindful of the nutritional contents of foods, and take charge of the food-preparation process.
- Keep goals in mind and post them prominently to stay motivated.